

Subject: Cloud ComputingDate: 10/4/25

• Sec-A (5x2=10)

1. What is security in cloud?

Ans cloud security refers to the set of policies, technologies and controls used to protect data, applications, and infrastructure in cloud environment from threats and unauthorized access.

2. What is Configuration Management?

Ans Configuration Management is the process of systematically handling changes to ensure a system maintains its integrity over time. In the cloud, it involves managing hardware, software and system settings to maintain consistency.

3. List top most common cloud security vulnerabilities:

Ans • Data breach

• Insecure APIs

• Misconfigured cloud settings

• Account hijacking

• Insider threats

4. Explain Security Management Standards.

Ans Security management standards are guidelines to protect information assets. Examples include:

• ISO/IEC 27001 - Information Security Management Systems (ISMS)

• NIST Framework - provides risk-based cybersecurity guidelines.

• HIPAA :- For healthcare data protection.

5. What is PaaS?

Ans Platform as a Service (PaaS) is a cloud computing model that provides a platform allowing developers to build, test, and deploy applications without managing the underlying infrastructure. ex: Google App Engine.

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• Section-B (3x7=21)

6. Explain PaaS availability management.
- Ans PaaS availability management ensures that the Platform Services provided to developers are continuously accessible, reliable and meet agreed performance levels.

Key Components: -

- (i) High Availability Architecture - cloud providers design Platform with redundancy across multiple data centres or regions to avoid downtime during failures.
- (ii) Auto-scaling - Automatically adjusts resources based on demand to maintain performance and availability during peak loads.
- (iii) Failover Mechanisms - if one component fails, the system switches to a backup component to ensure continuous operation.
- (iv) Service Level Agreements (SLAs): - PaaS providers typically offer SLAs that define the expected level of availability, uptime, and other performance metrics.

• Benefits of PaaS Availability Management: -

- (i) Reduced Downtime - Ensures applications and development tools are always accessible, reducing disruptions to business operations.
- (ii) Better User Experience - High availability means users and developers can access services without interruption leading to smoother app development and deployment.
- (iii) Increased Reliability - Failover systems and redundancy make the platform more dependable even during hardware or software failures.

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7. Explain the Key Privacy Concern in the cloud.

Ans The primary privacy concern in the cloud is data breach and unauthorized access. It may be misuse of personal or sensitive data stored and processed on third party servers.

Key Points:-

- (i) Data Confidentiality - cloud users worry about who can access their data. If data is not properly encrypted, it can be exposed to hacker or even unauthorized cloud provider staff.
- (ii) Data Control and Ownership - Users often lose direct control over their data once it is stored in the cloud, raising concerns about how it's used, shared or moved across regions.
- (iii) Data Breaches and Leaks - cloud environments are common targets for cyberattacks and breaches can result in sensitive information being leaked or stolen.
- (iv) Third-party access - cloud providers and their subcontractors may have access to user data, which increases the risk of data being used for purposes beyond the user's consent.

8. Explain SaaS availability management.

Ans SaaS (Software as a Service) availability management ensures that cloud-based software application are accessible, reliable and functional for users at all times. This involves monitoring, troubleshooting, and maintaining the infrastructure and software to ensure high uptime and prevent disruptions. SaaS providers are responsible

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for maintaining the software, applying updates and managing security to ensure high availability.

Key aspects of SaaS availability management -

- (i) Monitoring - Continuously tracking the performance and availability of SaaS applications, including uptime, response times and error rates.
- (ii) Troubleshooting - Identifying and resolving issues that impact availability, such as network problems, application errors, or server outages.
- (iii) Maintenance - Applying software updates, patching vulnerabilities and performing routine maintenance to ensure the stability and reliability of the SaaS application.
- (iv) Service Level Agreements (SLAs) - Establishing clear expectations regarding availability and performance, providing guarantees to users.
- (v) Disaster Recovery - Implementing plans to recover from major incidents, such as natural disasters or system failures, ensuring business continuity.

• Section-C (3x11=33)

9. Explain Access Control and Security vulnerability in cloud.

Ans Access Control refers to the methods used to regulate who can view or use resources in a cloud environment.

Key Elements -

- Authentication - Verifying the identity of users (e.g. password, MFA).
- Authorization - Granting users permission to access specific resources based on their roles.

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- Role-Based Access Control (RBAC) - Assigns access rights based on user roles to minimize unnecessary access.
- Least Privilege Principle - Users are given the minimum level of access required to perform their tasks.

Benefits - Reduced risk of security breaches, improved data protection and compliance with security standards.

(ii) Security Vulnerability in cloud -

Security vulnerabilities are weaknesses in a system that can be exploited by hackers to gain unauthorized access, steal data or disrupt services.

• Common Vulnerabilities:-

- (i) Insecure APIs - Poorly secured APIs can expose data to attackers.
- (ii) Data Breaches - Unauthorized access to sensitive data stored in the cloud.
- (iii) Misconfigured cloud settings - Leaving cloud storage open to the public or weak security settings.
- (iv) Account Hijacking :- Attackers gain access to cloud accounts through phishing or weak passwords.
- (v) Lack of Encryption - Data not encrypted in transit or at rest can be intercepted.

10. Explain security management in cloud computing in detail.
 Ans - Security management in cloud computing involves implementing policies, technologies and controls to protect cloud data, applications and infrastructure from threats and unauthorized access.

(i) Key Areas of Cloud Security Management -

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(a) Identity and Access Management (IAM): -

- Controls who can access cloud resources.
- Uses tools like Multi-factor Authentication (MFA), Role-Based Access control (RBAC) and Single sign-on (SSO).
- Ensures only authorized user access the right resources.

(b) Data Protection -

- Encryption - Data is encrypted in transit and at rest using security algorithms.
- Backups - Regular backups ensures data recovery in case of loss or attack.

(c) Network Security -

- Firewall and Intrusion Detection System (IDS) - Prevent unauthorized access and detect suspicious activities.
- VPNs - Secure communication channels over public networks.
- Segmentation - Divides network to isolate sensitive systems from potential threats.

(ii) Compliance and Governance -

- Ensures that cloud services comply with regulations like GDPR, HIPAA, ISO/IEC 27001 and NIST.
- Regular audits and assessments are conducted to maintain legal and security standards.

(iii) Monitoring and Incident Response -

- Continuous monitoring of cloud resources for security breaches.
- Well defined incident response plans helps mitigate damage quickly during attacks.

(iv) Physical Security - Cloud data centres use physical security measures like surveillance, access control, and environmental controls to prevent unauthorized access and damage.

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31. Explain Patch and Configuration Management.

Ans. Patch management is the process of identifying, acquiring, testing and deploying updates (patches) to software and firmware to address vulnerabilities, fix bugs or improve functionality.

Key activities:-

- Vulnerability Scanning - Detect system that require updates.
- Patch Deployment - Apply updates regularly to fix known issues.
- Testing - Ensure patches don't cause conflicts or failures.
- Automation - Use tool to automate patching across cloud systems.

Benefits:

- Fixes security flaws, reducing the risk of attacks.
- Improves system stability and performance.
- Ensures compliance with security standards.

(ii) Configuration Management →

Configuration Management involves systematically managing system settings, software versions and hardware configurations to ensure consistency and security across cloud environments.

Key Activities:-

- Baseline Configuration - Establish standard configurations for cloud resources.
- Change Control - Track and authorize changes to avoid errors.
- Automation Tools - Tools like Ansible, Puppet and Chef manage configurations at scale.
- Monitoring → Detect and correct unauthorized changes.

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Benefits:

- ↳ Maintains uniformity across cloud systems.
- ↳ Reduces human errors and misconfigurations.
- ↳ Enhances scalability, reliability and security.

• Relationships:—

Patch and Configuration management are often integrated within a larger IT change management process.
